Tree Potentials and Path Costs: Quantitative Stopping, Predictable Energy, and Norm-Faithful Ram Certificates

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

We study the exact real Banach norms in the source manuscript *Midpoint convexity from bounded tree potentials and path costs*. The main local refinement retains the quadratic quotient cost when stopping signed root-path mass. It bounds an arbitrary displacement outside a finite ancestral head by K_* times the square root of the average squared endpoint excess, where $K_* < 52/17 < 7$. The bound yields explicit separated-lens and approximate-head estimates. A weighted subtree-quota argument also extends the source's dyadic predictable-layer estimate to finite filtrations with child probabilities bounded below; the dyadic constant becomes 25, rather than 252. An exact subspace identification transfers the already established averaged midpoint estimate to the free-root edge model. Finite Ram representations preserve the actual mixed Hilbert and absolute-mass cost, not an invented Euclidean metric. Classical convexity, stopping, duality and conditional expectation are credited, while the source's global renorming statements and companion results receive no new independent certification from our finite tests. Contribution assessment: C3/5, explicit quotient-preserving stopping and weighted predictable-energy refinements give notable local improvements without establishing a new global renorming theorem.

1 Models and logical scope

All scalars are real. Let $T_n = \mathbb{N}^{\le n}$, with root o, and order vertices by ancestry. Different n's carry distinct copies of the tree. Put $H_n = \ell_2(T_n)$,

$$b_t = \sum_{s \preceq t} e_s, \qquad J_n\lambda = \sum_t \lambda_t b_t, \qquad \lambda \in \ell_1(T_n).$$

The series converges absolutely in H_n, since $\|b_t\|_2 \le \sqrt{n+1}$. The source [1] uses two different quotient norms:

$$L_n(z) = \inf_{z=h+J_n\lambda} \big(\|h\|_2 + \|\lambda\|_1\big), \tag{1}$$

$$Q_n(z) = \inf_{z=h+J_n\lambda} \big(\|h\|_2^2 + \|\lambda\|_1^2\big)^{1/2}. \tag{2}$$

Infima need not be attained. Their outer Hilbert sums are denoted $X_L = (\bigoplus_n (H_n, L_n))_2$ and $X_q = (\bigoplus_n (H_n, Q_n))_2$. Their dual block norms are, respectively,

$$\max\{\|f\|_2, \sup_t |\langle b_t, f\rangle|\}, \qquad \left(\|f\|_2^2 + \sup_t |\langle b_t, f\rangle|^2\right)^{1/2}.$$

These follow by pulling back a functional to $H_n \oplus_1 \ell_1(T_n)$ or $H_n \oplus_2 \ell_1(T_n)$; testing Hilbert vectors and single path atoms proves the reverse bounds. In particular $Q_n \le L_n \le \sqrt{2} Q_n$, but $L_n(e_o) = 1$ and $Q_n(e_o) = 1/\sqrt{2}$. Uniform equivalence is not equality of midpoint constants.

A finite *ancestral head* A contains every ancestor of each retained node. Its coordinate restriction is $P = P_A$, and $Q = I - P$. Let $r_A(t)$ be the last ancestor of t in A, or a dummy symbol if there is none. Fold endpoint masses by

$$(R_A \lambda)_r = \sum_{r_A(t) = r} \lambda_t,$$

discarding the dummy fiber. Absolute convergence gives $\|R_A \lambda\|_1 \le \|\lambda\|_1$ and $PJ = JR_A$. Thus P is contractive in both models, and $\|Q\| \le 2$. Neither complementary projection is silently assumed contractive.

For the quadratic model use the genuine global spaces

$$H = (\bigoplus_n H_n)_2, \qquad \mathcal{L} = (\bigoplus_n \ell_1(T_n))_2.$$

The global formula is

$$\|z\|_q^2 = \inf_{z = h + J\lambda} (\|h\|_H^2 + \|\lambda\|_{\mathcal{L}}^2). \tag{3}$$

Choose component representations with summable squared errors to prove this identity. The map $J : \mathcal{L} \to X_q$ is bounded by one; there is no uniform assertion $J : \mathcal{L} \to H$. Each finite-height block is Hilbert-equivalent and reflexive. Its constants depend on height; reflexivity of the outer sum is a standard consequence, not a uniform Hilbert representation of that sum.

The source also contains four infinite-height potential norms, a free-root edge norm, a start-charged segment norm and an incomparable-segment dual. Sections 5 and 7 distinguish them. Our main theorem concerns precisely (2), while the predictable-energy theorem concerns precisely (1). No theorem here substitutes a finite Gram condition for the existence of compatible infinite tests or weakly null trees.

2 Retaining the quadratic stopping cost

Suppose $Px = x$, and choose endpoint representations

$$x \pm y = h^{\pm} + J\lambda^{\pm}.$$

Define

$$\bar{h} = (h^+ + h^-)/2, \quad \widetilde{h} = (h^+ - h^-)/2, \quad \sigma = (\lambda^+ + \lambda^-)/2, \quad \tau = (\lambda^+ - \lambda^-)/2.$$

In block n, let

$$m_n = (|\lambda_n^+| + |\lambda_n^-|)/2, \quad M_n = \|m_n\|_1, \quad q_n = \|R_A \sigma_n\|_1, \quad d_n = M_n - q_n.$$

Here $|\sigma_n|, |\tau_n| \le m_n$ pointwise and $d_n \ge 0$. Let C be the average of the two squared representation costs. The source energy identity, whose short proof we retain, gives

$$\left\|Q\bar{h}\right\|_H^2 + \left\|\widetilde{h}\right\|_H^2 + \sum_n d_n^2 \le C - \|x\|_q^2. \tag{4}$$

Indeed the parallelogram identity and scalar convexity imply $C \geq \left\|\bar{h}\right\|_H^2 + \left\|\widetilde{h}\right\|_H^2 + \sum M_n^2$. The representation $x = P\bar{h} + JR_A\sigma$ has squared cost $\left\|P\bar{h}\right\|_H^2 + \sum q_n^2 \geq \|x\|_q^2$. Subtract, and use $M_n^2 - q_n^2 \geq (M_n - q_n)^2$. This is a shared energy budget; the three terms must not each be paid the entire budget and then combined linearly.

Lemma 2.1 (Parameterized signed stopping)**.** *Fix $0 < \theta < 1$. There is a splitting $\tau = \tau^{\mathrm{r}} + \tau^{\mathrm{k}}$ such that*

$$\|\tau_n^{\mathrm{r}}\|_1 \leq d_n/(1-\theta), \tag{5}$$

$$QJ\tau^{\mathrm{k}} \in H, \qquad \left\|QJ\tau^{\mathrm{k}}\right\|_H \leq \theta^{-1}\left\|Q\bar{h}\right\|_H. \tag{6}$$

It is valid for countable endpoint support and countably many children at every finite height.

Proof. For a node $t \notin A$, let S_t be its descendant set. Mark t if $|\sigma_n(S_t)| < \theta m_n(S_t)$, and retain only the first marked node on each exterior path. Every node has finitely many ancestors, so the first-marked sets are disjoint. They lie within fibers of r_A. For disjoint subsets S_j of a nondiscarded fiber B,

$$\sum_j (m_n(S_j) - |\sigma_n(S_j)|) \leq m_n(B) - |\sigma_n(B)|.$$

To see this, bound $|\sigma_n(B)|$ by the absolute signed masses of the subsets plus the m_n-mass of their complement. Countable subsets are allowed by absolute convergence. For discarded fibers use their whole mass. Summing yields

$$(1-\theta) \sum_{\text{first marked } t} m_n(S_t) \leq d_n.$$

Let τ_n^{r} be the restriction of τ_n to that union, and keep the complement. This proves (5). At an exterior node with a marked ancestor the kept occupancy is zero. At any other exterior node it is at most $m_n(S_t)$, hence at most $\theta^{-1}|\sigma_n(S_t)| = \theta^{-1}|\bar{h}_n(t)|$, because $Qx = 0$. Square and sum the coordinate inequalities. The resulting Hilbert array agrees coordinatewise with $QJ\tau^{\mathrm{k}}$. Coordinate continuity and the natural inclusion $H \to X_q$ identify the two elements, proving (6). □

Theorem 2.2 (Optimized quotient-tail bound)**.** *For every finite ancestral head A, every $x, y \in X_q$ with $P_A x = x$, and every $0 < \theta < 1$, put*

$$K_\theta^2 = \max\{1 + \theta^{-2},\, 4/(1-\theta)^2\}, \qquad D(x,y) = \frac{\|x+y\|_q^2 + \|x-y\|_q^2}{2} - \|x\|_q^2.$$

Then $D(x,y) \geq 0$, and

$$\|Q_A y\|_q^2 \leq K_\theta^2 D(x,y). \tag{7}$$

The best constant within this scalar threshold majorant is attained at

$$\theta_* = \frac{1 + \sqrt{5} - \sqrt{2 + 2\sqrt{5}}}{2}, \tag{8}$$

$$K_* = \frac{2}{1-\theta_*} = \sqrt{1 + \theta_*^{-2}} < \frac{52}{17} < 7. \tag{9}$$

The assertion is not that K_* is the optimal geometric constant of the Banach space.

Proof. Use endpoint representations whose average squared cost is at most the average squared endpoint norm plus $\eta > 0$. Write

$$a = \left\|Q\widetilde{h}\right\|_H, \quad b = \left\|Q\bar{h}\right\|_H, \quad c = (\sum_n d_n^2)^{1/2}.$$

By (4), $a^2 + b^2 + c^2 \le D(x, y) + \eta$. Crucially, the following is one admissible representation in (3), not a sum of three unrelated norm bounds:

$$Qy = \underbrace{Q\widetilde{h} + QJ\tau^{\mathrm{k}}}_{\text{Hilbert part}} + J\underbrace{(\tau^{\mathrm{r}} - R_A\tau^{\mathrm{r}})}_{\text{endpoint part}}. \tag{10}$$

Its cost squared is at most

$$(a + b/\theta)^2 + \frac{4c^2}{(1-\theta)^2} \le K_\theta^2(a^2 + b^2 + c^2).$$

The factor two for the endpoint part is justified by the $\mathcal{L}$-contractivity of R_A. In components with empty head it can be replaced by one. Let $\eta \downarrow 0$. Convexity already gives $D \ge 0$.

The first entry of K_θ^2 decreases strictly and the second increases strictly. Their unique crossing solves

$$(1 + \theta^2)(1 - \theta)^2 = 4\theta^2.$$

After division by θ^2, put $u = \theta + \theta^{-1}$. The equation becomes $u^2 - 2u - 4 = 0$; $u > 2$ selects $1 + \sqrt{5}$, giving (8). At $\theta = 9/26$,

$$1 + \theta^{-2} = 757/81 < 2704/289 = 4/(1 - \theta)^2.$$

The unique optimizer therefore has $K_* < 52/17$. □

Remark 2.3 (What changed). The source's stopping proof fixes $\theta = 1/2$ and concludes $\|Qy\|_q \le a + 2b + 4c \le 7\sqrt{D}$. Retaining joint energy alone would give $\sqrt{21}$. Retaining the actual quotient representation gives 4 at the same threshold, and optimizing the threshold gives K_*. No new stopping principle, Hilbert norm on $\mathcal{L}$, or optimal renorming modulus is claimed.

2.1 Data-dependent certificates

Before using the scalar majorant, a supplied exact representation gives

$$\|Qy\|_q^2 \le \left(a + \theta^{-1}b\right)^2 + \sum_n \|\tau_n^{\mathrm{r}} - R_A\tau_n^{\mathrm{r}}\|_1^2. \tag{11}$$

This can be smaller than (7); it retains cancellation during folding and the actual removed sets. An upper bound on representation cost and a certified lower bound on $\|x\|_q$ suffice for an upper bound on its energy deficit. A finite norm optimizer alone does not certify an infinite endpoint support or an unmeasured tail.

3 Heads, lenses and quantitative consequences

Corollary 3.1 (Approximate head and unequal endpoints)**.** *For $x, y \in X_q$, a finite ancestral head A, and $\rho = \|x - P_A x\|_q$, let $R_2 = ((\|x+y\|_q^2 + \|x-y\|_q^2)/2)^{1/2}$. Then*

$$\|Q_A y\|_q \le K_* \sqrt{(R_2+\rho)^2 - \|P_A x\|_q^2}. \tag{12}$$

In particular, if $\|x \pm y\|_q \le (1+\delta)\|x\|_q$, $x \neq 0$, and A is selected so that $\rho \le \varepsilon \|x\|_q$, $0 < \varepsilon < 1$, then

$$\|Q_A y\|_q \le K_* \|x\|_q \sqrt{(1+\delta+\varepsilon)^2 - (1-\varepsilon)^2}.$$

The same head works for every such displacement y.

Proof. Apply Theorem 2.2 to $P_A x, y$. Minkowski's inequality for the two scalar endpoint norms bounds their root mean square by $R_2 + \rho$. Also $\|P_A x\|_q \ge \|x\|_q - \rho$. Strong convergence of finite head restrictions permits the stated choice of A before y is specified. □

Corollary 3.2 (Separated symmetric lenses)**.** *If $\|x \pm y_j\|_q \le R$ and $\|y_j - y_k\|_q \ge \epsilon > 0$ for all distinct indices in an infinite sequence, then*

$$\|x\|_q^2 + \frac{\epsilon^2}{4K_*^2} \le R^2.$$

Proof. Approximate x by $P_A x$ with error ρ. Corollary 3.1 bounds all tails by $K_* \sqrt{(R+\rho)^2 - \|P_A x\|_q^2}$. The bounded head projections lie in a finite-dimensional space, so two can be arbitrarily close. Their separation is at most twice the tail bound plus this arbitrarily small head distance. Let $\rho \downarrow 0$. □

For a finite supported unit center and $y \in \ker P_A$ with $\|y\|_q \ge 1$, (7) implies $\max_\pm \|x \pm ty\|_q \ge \sqrt{1 + t^2/K_*^2}$. Both endpoint norms are at least one by contractivity of P_A. Approximating a general unit center with arbitrary accuracy, before selecting y, therefore gives

$$\delta_{X_q}^{\max}(t) \ge \sqrt{1 + t^2/K_*^2} - 1, \qquad \widehat{\delta}_{X_q}(t) \ge \frac{\sqrt{1 + t^2/K_*^2} - 1}{2}. \tag{13}$$

Here the moduli have the order $\inf_{\|x\|=1} \sup_{\operatorname{codim} F < \infty} \inf_{y \in F, \|y\| \ge 1}$, using respectively the maximum or average of the two endpoint norms, minus one. These estimates are for X_q; the already known $1/4$ coefficient for the four potential models is not being claimed again.

4 Weighted predictable layers in the linear model

This section uses X_L, not X_q. A finite initial set T in one block has the exact path representation

$$P_T z = \sum_{u \in T} \left(z(u) - \sum_{\substack{v \text{ child of } u \\ v \in T}} z(v) \right) b_u.$$

Denote the sum of absolute values of these coefficients by $V_T(z)$, and set $V_\varnothing = 0$. Folding a path representation gives $V_T(J_n \lambda) \le \|\lambda\|_1$. An interval cut from a root path has L_n-norm at most two. These are actual signed-mass statements; arbitrary coordinatewise domination does not imply domination in L_n.

Lemma 4.1 (Weighted mass removal)**.** *Let $S \subseteq T \subset T_n$ be finite initial sets, and $z = \sum_{j=1}^{r} \pi_j z_j$, where $\pi_j > 0$ and $\sum_j \pi_j = 1$. Put $U = T \setminus S$ and*

$$g = \sum_j \pi_j V_T(z_j) - V_S(z).$$

Then $g \geq 0$, and for each j,

$$L_n(P_U z_j) \leq \pi_j^{-1} \|P_U z\|_2 + 2\pi_j^{-1} g. \tag{14}$$

Proof. If $S = \varnothing$, then $L_n(P_T z_j) \leq V_T(z_j) \leq g/\pi_j$. Suppose $S \neq \varnothing$. Collapse the coefficients of each $P_T z_j$ to their last ancestor in S. Orient each fiber by the sign of the coefficient of $P_S z$, choosing either sign at zero. Write the oriented refined coefficients as $p_u^j - q_u^j$, with $p_u^j, q_u^j \geq 0$ and disjoint signs. Let bars denote weighted averages over j. The collapse identity gives

$$g = 2 \sum_{u \in T} \bar{q}_u.$$

In an exterior subtree all descendants have the same orientation. Writing $P^j(v), Q^j(v), \bar{P}(v), \bar{Q}(v)$ for descendant sums,

$$P^j(v) \leq \pi_j^{-1} \bar{P}(v) \leq \pi_j^{-1}(|z(v)| + \bar{Q}(v)).$$

We use the source's subtree-quota construction, with its proof included. For nonnegative endpoint masses μ, ν on a finite forest set $q(v) = \min\{\mu(S_v), \nu(S_v)\}$. Then $\sum_{w \text{ child of } v} q(w) \leq q(v) \leq \mu(S_v)$. Working upward from the leaves, mark enough unmarked μ-mass to meet $q(v)$; fractions are allowed. Later ancestor steps can only increase an already met quota. The total marked mass in each tree is its root quota and is at most the total ν-mass. Thus a submass $\rho \leq \mu$ meets every quota with $\|\rho\|_1 \leq \|\nu\|_1$.

Apply this to the positive endpoints p^j in U, with comparison mass $\pi_j^{-1}\bar{q}$. The removed positive mass is at most $g/(2\pi_j)$. The kept positive occupancy at v is at most

$$(P^j(v) - \pi_j^{-1}\bar{Q}(v))_+ \leq \pi_j^{-1}|z(v)|.$$

Its Hilbert norm has the same bound. The negative child mass is also at most $g/(2\pi_j)$. Each removed or negative exterior path costs at most two. This gives (14). □

Theorem 4.2 (Predictable energy with nonuniform branching)**.** *Let $(\mathcal{F}_i)_{i=0}^m$ be a finite probability filtration, each $\mathcal{F}_i$ generated by a finite atomic partition, with trivial $\mathcal{F}_0$. On each positive-probability parent atom, every child has conditional probability at least $\eta > 0$. Let $x_i = \mathbb{E}[x_m \mid \mathcal{F}_i]$ take values in X_L. Let A_0 be a fixed finite initial set and A_i be finite initial sets measurable with respect to $\mathcal{F}_{i-1}$, nested on every history, with* $\operatorname{supp} x_i \subseteq A_i$*. For $\Delta_i = A_i \setminus A_{i-1}$,*

$$\sum_{i=1}^{m} \mathbb{E}\,\|P_{\Delta_i} x_i\|_{X_L}^2 \leq \left((1+\eta^{-1})^2 + 4\eta^{-2}\right) \mathbb{E}\,\|x_m\|_{X_L}^2 . \tag{15}$$

For fair binary branching the coefficient is 25. No depth or block-height constant is hidden in this estimate.

Proof. Fix $\zeta > 0$. On each terminal outcome choose component representations

$$x_m^n = h_n + J_n \lambda_n, \qquad \|h_n\|_2 + \|\lambda_n\|_1 \leq (1+\zeta) L_n(x_m^n).$$

Use zero in a zero component. Projecting to the finite terminal head retains the representation and decreases both costs. There are finitely many histories, so all ensuing arrays can be taken on a common finite set. Put $h_i = \mathbb{E}_i h$, $b_i = (J_n \mathbb{E}_i \lambda_n)_n$, and

$$H_0^2 = \mathbb{E} \sum_n \|h_n\|_2^2 , \qquad M_0^2 = \mathbb{E} \sum_n \|\lambda_n\|_1^2 .$$

The joint budget, rather than two independent upper bounds, is

$$H_0^2 + M_0^2 \le (1+\zeta)^2 \mathbb{E} \|x_m\|_{X_L}^2 . \tag{16}$$

On a parent atom apply Lemma 4.1 to the children of b_i^n, with $S = A_{i-1} \cap T_n$ and $T = A_i \cap T_n$. These sets are fixed before the new outcome. If $g_{i,n}$ is the nonnegative variation gain, conditional Jensen gives

$$V_{A_{i-1}\cap T_n}(b_{i-1}^n)^2 + g_{i,n}^2 \le \mathbb{E}_{i-1} V_{A_i \cap T_n}(b_i^n)^2 .$$

Telescoping yields $\sum_{i,n} \mathbb{E} g_{i,n}^2 \le M_0^2$. The support condition gives $P_{\Delta_i} b_{i-1} = -P_{\Delta_i} h_{i-1}$. Thus

$$\|P_{\Delta_i} x_i\|_{X_L} \le \|P_{\Delta_i} h_i\|_H + \eta^{-1} \|P_{\Delta_i} h_{i-1}\|_H + 2\eta^{-1} \left(\sum_n g_{i,n}^2 \right)^{1/2} .$$

For $j = i-1, i$, predictability allows $P_{\Delta_i} h_j = \mathbb{E}_j (P_{\Delta_i} h)$. Conditional Jensen and pathwise disjointness of the layers bound each of the two unscaled Hilbert energies by H_0^2. Minkowski in the scalar L_2 space of time and outcomes therefore bounds the square root of the left side of (15) by

$$(1+\eta^{-1}) H_0 + 2\eta^{-1} M_0 \le \sqrt{(1+\eta^{-1})^2 + 4\eta^{-2}} \sqrt{H_0^2 + M_0^2} .$$

Use (16) and let $\zeta \downarrow 0$. □

The source's variation proof uses two-approximate terminal representations and separates the three square budgets, obtaining $3(1+4+16)4 = 252$. The gain to 25 uses arbitrarily near-optimal representations, the shared terminal budget and scalar Minkowski. The extension to unequal child probabilities additionally requires Lemma 4.1. Neither 25 nor its probability dependence is claimed optimal. As $\eta \downarrow 0$ the bound deteriorates; one cannot silently condition on arbitrarily rare events. If the parent has mass in its new window, the vanishing identity in the proof fails. The source's separate leakage theorems are then required. Similarly, heads chosen after seeing the child need not commute with parent conditional expectation.

5 Free-root identification and exact norm examples

On $T = \mathbb{N}^{<\omega}$, the root-inclusive sibling model has tests

$$|P_f(t)| \le 1, \qquad \sum_{w \text{ child of } v} |f_w|^2 \le 1, \qquad P_f(t) = \sum_{s \preceq t} f_s .$$

The root is controlled by $|P_f(o)| \le 1$; it is not a child in the quadratic constraint. Its norm is the supremum of $\sum_s x_s f_s$. In the free-root edge model, tests are potentials $b : T \to [-1,1]$ with $\sum_{w \text{ child of } v} |b_w - b_v|^2 \le 1$, evaluated as $\sum_{s \ne o} x_s (b_s - b_{s^-})$. The root value is free but has no edge coordinate.

Proposition 5.1 (An exact closed subspace)**.** *The free-root edge space is isometric to the kernel of the root coordinate functional in the completed root-inclusive sibling space. Consequently it inherits the source estimate*

$$\widehat{\delta}_{\mathrm{edge}}(t) \geq \sqrt{1 + t^2/4} - 1, \qquad 0 < t < 1.$$

Proof. Set $f_o = b_o$ and $f_s = b_s - b_{s^-}$ for $s \neq o$. The two test sets correspond bijectively by telescoping. For vectors with zero root coordinate the evaluations agree exactly. This isometry extends to completions. Its image is the full kernel: approximate a kernel vector by finite arrays, then subtract their vanishing root coordinates times e_o. For any closed subspace Y of a Banach space, a finite-codimensional subspace F intersects Y in finite codimension in Y. The inherited norm and the smaller set of directions imply that the averaged modulus of Y is at least the ambient one. Apply the source's already proved root-inclusive estimate. □

This simplifies and quantitatively strengthens the source's separate planar free-root bound $(\sqrt{1 + t^2/576} - 1)/4$, but is only an isometric identification and inheritance argument, not a new midpoint-convexity mechanism. It does not identify the free-root norm with the model fixing $b_o = 0$. For a path with two edges, the sum of its edge vectors has free-root norm two: choose labels $-1, 0, 1$ on that path and freeze off-path branches. The sibling bounds are one, and telescoping gives two. With root fixed at zero the same norm is one. Both upper bounds also follow from telescoping. In the root-inclusive space a whole root path has norm one, whereas deleting its root can have norm two; hence the complement of the root projection has norm exactly two.

5.1 A quadratic block is not a Euclidean coefficient space

On the ancestral two-node head $\{o, v\}$, the columns of J are $(1, 0)^T, (1, 1)^T$. The following pairs $z = h + J\lambda$ are exact:

z	h	λ	$Q_n(z)^2$
$(1, 0)$	$(1/2, 0)$	$(1/2, 0)$	$1/2$
$(0, 1)$	$(-1/3, 2/3)$	$(0, 1/3)$	$2/3$
$(1, 1)$	$(1/3, 1/3)$	$(0, 2/3)$	$2/3$
$(1, -1)$	$(1/2, -1)$	$(1/2, 0)$	$3/2$

For each row the displayed squared cost equals $h \cdot z$ and

$$Q_n^*(h)^2 = h_o^2 + h_v^2 + \max\{h_o^2, (h_o + h_v)^2\}.$$

Duality therefore proves equality, not just a feasible upper bound. Extension by zero of the dual array and folding of paths show that these are the actual infinite-branching block norms. The parallelogram sides for e_o, e_v are $13/6$ and $7/3$, respectively. Thus no positive-definite Gram matrix gives this norm on even this two-dimensional head.

6 Ram representations with the cost retained

The original compatible-panel proof [6], specifically its coordinate-transfer theorem and constrained-energy section, keeps a physical space fixed and transports both operators and its norm. That principle applies here, but its Hilbert mass-matrix formula alone does not describe a path quotient.

On a finite head take invertible synthesis maps U, V, with physical arrays $h = Uu$, $\lambda = Vv$. The exact quadratic cost and output become

$$z = Uu + JVv, \qquad u^T U^T U u + \|Vv\|_1^2 . \tag{17}$$

The linear cost is $(u^T U^T U u)^{1/2} + \|Vv\|_1$. The absolute mass term is not generally $v^T G v$ for a single matrix G. Its expression changes with the sign cone. All descendant occupancies and folding operations must be applied to the physical coefficients Vv; unrelated coordinates on overlapping paths do not specify a shared measure.

For example take $V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then $\|Vv\|_1^2 = (|v_1 + v_2| + |v_2|)^2$. Take physical endpoint vectors $Vv = (1, 0)$, $Vw = (0, 1)$. Their squared costs are $1, 1, 4, 4$; the parallelogram identity would require $8 = 4$, an exact failure.

The local original [7] gives a finite conditional coupling criterion with all shared-state masses and exact interval constraints. Its relevant lesson here is to preserve the shared fiber and positivity conditions. It is not a Banach-space renorming proof and does not supply countable-support stopping or finite-codimensional selection. The later comparison [8] bounds residual errors in an exact polynomial identity; we use only its insistence on keeping an error budget instead of treating a small residual as zero.

For an inexact finite representation $z = h + J\lambda + e$,

$$Q_n(z) \le \sqrt{\|h\|_2^2 + \|\lambda\|_1^2} + \|e\|_2 .$$

A dual-feasible f gives the lower bound $Q_n(z) \ge f(z)$. Thus a valid finite certificate consists of an output identity or explicit residual, the actual mixed cost, and every path constraint of the dual witness. A lower bound on center norm can then enter (4). Approximate stopping tests require certified inequalities for both signed occupancies and positive mass. For example,

$$|\widehat{\sigma}(S)| + \epsilon_\sigma < \theta\big(\widehat{m}(S) - \epsilon_m\big)$$

certifies a mark when the errors bound the respective true quantities. An ambiguous test is not an exact stopping decision. No truncation to a finite head proves that an omitted infinite coefficient tail is small unless that tail is separately bounded in $\mathcal{L}$ and, where needed, in H.

7 Source architecture and retained obligations

The cached manuscript has seven scientific section files, a main file, a figure source, a bibliography and a README. All were literally read in bounded displays for this report; reading is not a claim that every global statement has been independently reproved.

Module	Mechanism and treatment here
Introduction	Distinguishes averaged and maximum midpoint moduli, one-sided AUC, finite-height blocks and infinite completions.
Constrained	Four norming sets: sibling constraints with/without root, antichain square constraints, and global ℓ_2 constraint. The paired relative-potential proof and bounded weakly null tree obstruction are source results, not new contributions here.

Module	Mechanism and treatment here
Selections	One-part, four-part and independent-component choices retain different witness data. Their weaker numerical bounds are not needed for our quadratic stopping theorem.
Free root	Scalar bounded potentials and planar contraction construction. Proposition 5.1 supplies a simpler subspace transfer for this specific norm, without modifying the source's valid witnesses.
Linear	Exact additive cost, dual clipping, positive-mass monotonicity, subtree quotas, stopped fibers, and predictable/adaptive-window energy. We reprove the weighted quota step and retain a joint terminal budget; the separate leakage results remain source inputs.
Quadratic	Exact $H \oplus_2 \mathcal{L}$ quotient, folding deficit, stopped flows, clipping and three cancellation decompositions. Theorem 2.2 refines its signed stopping route and uses the average squared endpoint cost directly.
Segments	Start masses combine additively along branches and quadratically across children; cancellation is charged at the later start. The incomparable-segment norm requires nodewise incomparability, not just disjoint supports. Neither norm is identified with X_q.

The two segment constructions retain their source constants $1/64$ and $\sqrt{2}+\sqrt{6}$ in their respective midpoint/tail statements. We do not claim a transfer of K_* to them. The source's weakly null children live in fixed finite-height blocks or use its actual dual coefficient restrictions. Finite rational arrays cannot prove weak nullity against an entire infinite-dimensional dual. Likewise, bounded trees of arbitrary height, not one large finite tree, are required for the no-AUC-renorming argument.

The family companions M638 and M643 concern inherited averaged potential and recursive-potential estimates. Their constants and global conclusions are not part of this report's earned increment. Our improvements are in the quadratic root-path stopping budget and the linear path-variation budget, with a minor free-root identification. We did not rebuild a Lean development or independently certify the whole family.

8 Current literature and comparison

The public-source refresh was performed on October 9, 2026, using only public mathematical titles and URLs. The original repository introduction was located publicly; the technical work uses the pinned local text. The captured-tree identity supplied for the batch is provenance, not a statement about a live branch or a proof-validity certificate.

Dilworth et al. [2] introduce the maximum form of AMUC and relate it to the noncompactness of symmetric lenses. We consulted the author manuscript's definitions and Theorem 2.1 with its proof. This supplies historical context, not a new attribution for finite-rank head approximation or lens separation. The diamond-graph paper [3] has an unconditional-asymptotic-structure hypothesis in its reflexive characterization; its abstract and publication metadata were refreshed, not its full proof. That hypothesis cannot be discarded by a finite tree calculation.

Baudier's September 18, 2026 preprint [4] states that the Kadets–Werner modification is AMUC but has no equivalent AUC norm. Its abstract and version record were checked. This already answers the general renorming question negatively; it should not be confused with an independent validation of the source's particular reflexive models. Perreau's public record [5] concerns bundle graphs

and dual spaces. It is contextual only. No historical priority claim is made for our quantitative corollaries.

9 Advantages, costs and further work

The quotient-tail estimate improves a finite geometric constant while allowing arbitrary head displacement, unequal endpoint norms and countable endpoint arrays. It uses the precise cost already present in the source; preserving that cost is the reason it beats a linear sum of separate error bounds. Its price is full control of the signed measure, folding map and the joint energy deficit. The scalar threshold is optimized, but the decomposition may still be geometrically wasteful. In particular the bound two on the folded removed mass can be strict.

The predictable-energy extension handles finite nonuniform probability trees and gives a concrete constant, but requires a common head selected before the child and exact support of the parent in its earlier head. It is not a result for arbitrary stochastic processes, arbitrary coordinate masks, or post-selected heads. The $1/\eta$ costs expressly record rare-child amplification. Near-optimal representations prove the bound without providing an efficient method for finding them.

The Ram formulation makes exact synthesis, shared path variables and residual budgets explicit. It does not simplify all sign cones, remove the growing number of tree constraints, or establish numerical speedups. No experiments measuring solver time were performed. An informative next step is to minimize the actual data-dependent bound (11) over thresholds on each block, with a certified common energy budget. Another is to derive a leakage version of Theorem 4.2 with probability-dependent weights rather than a worst-case η. Both require fresh proofs; neither follows merely by inserting a different constant into this theorem.

10 Evidence and conclusion

The literal-read ledger records before/after hashes and exact displayed intervals for all 4,302 lines of the eleven cached source text files. Private scope is narrower: TR250 lines 116–280 and 622–723; the local conditional-coupling report lines 1–250 and 270–410; and the later TR-item-047 comparison lines 299–391. The latter two are comparisons, not substitutes for the original compatible-coordinate and constrained-energy proofs in TR250. All complementary intervals remain explicitly unread. No included private references or entire confidential corpus are claimed read.

The owned executable checks passed 246 exact cases and 5,729 assertions, with zero floating-point diagnostic cases. They test actual signed path arrays, quotient representations, exact two-node norms, weighted quota feasibility and coefficient identities. They distinguish exact cases and assertions from numerical diagnostics, and exclude all independent external review checks. They supplement the proofs, especially their countable-sum and probability-quantifier steps; they do not certify global renorming. The delivered evidence also records compiler diagnostics, individual page inspection and final artifact hashes.

The substantive conclusion is quantitative and model-specific. Retaining an admissible quadratic representation improves signed stopping from 7 to $K_* < 52/17$, and a weighted removal proof with a shared cost budget gives 25 for dyadic predictable energy. These gains justify C3 as a notable local contribution; they do not claim new classical principles, optimal Banach-space constants, or an independent proof of all source and companion conclusions.

References

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[3] F. P. Baudier et al. *On the geometry of the countably branching diamond graphs*. J. Funct. Anal. 273 (2017), 3150–3199. arXiv:1612.01984; abstract and publication record consulted, full proof not audited.

[4] F. P. Baudier. *The Kadets–Werner modification of Bourgain–Rosenthal space is asymptotically midpoint uniformly convex*. arXiv:2609.21283v1, September 18, 2026; abstract and version record consulted.

[5] Y. Perreau. *On the embeddability of countably branching bundle graphs into dual spaces*. arXiv:2104.10494; public abstract/version metadata only.

[6] M. Subbarao. *Compatible Variational Ram Panels*, original TR250 manuscript, local confidential corpus. Coordinate transfer and constrained-energy proofs at the intervals stated in Section 10.

[7] M. Subbarao. *Ram Boolean Moments and Conditional Gluing for Completed Three-Edge Cycles*, TR-meta4-math, October 3, 2026. Local comparison, including its actual shared-state coupling proof.

[8] M. Subbarao. TR-item-047-astra, local comparison manuscript. Exact moment conditions and residual-aware identity proof, lines 299–391; not an independent tree-space theorem.